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15.1 Properties of Variance

Recall that the ***variance*** of a random variable X is

$$\text{Var}(X) = \mathbb{E}\left[(X-\mu)^2\right]$$

where $\mu = \mathbb{E}[X]$.

For a discrete random variable X, this expectation can be written as

$$\text{Var}(X) = \sum_x (x-\mu)^2 \mathbb{P}[X = x]$$

by Theorem 60.

The positive square root of the variance is called the ***standard deviation***.

There is a shortcut for computing variance, given in the next theorem.

Theorem 64 (Variance shortcut formula)**.** *Let X be a random variable with $\mathbb{E}\left[X^2\right] < \infty$. Then*

$$\text{Var}(X) = \mathbb{E}\left[X^2\right] - (\mathbb{E}[X])^2$$

The condition $\mathbb{E}\left[X^2\right] < \infty$ in Theorem 64 is there to exclude the possibility that the right hand side of the formula be the undefined form $\infty - \infty$.

While variance does give us some information, it's only a single number and so the amount of information it gives us about the random variable is actually very limited, and it is impossible to interpret it precisely without additional information about the random variable. This is illustrated in the following example.

Example 65. Let $M = 1,000,000$. And let

$$X = \begin{cases} 0 & : \text{ with probability } .99 \\ 10M & : \text{ with probability } .01 \end{cases}$$

The random variable X is like winning the lottery.

On the other hand, let

$$Y = \begin{cases} -10 & : \text{ with probability } \frac{1}{2} \\ 10 & : \text{ with probability } \frac{1}{2} \end{cases}$$

In the case of X, we have

$$\begin{aligned}
\text{Var}(X) &= \mathbb{E}\left[X^2\right] - (\mathbb{E}[X])^2 \\
&= \left[(10M)^2(0.01) + 0^2(0.99)\right] - \left[10M(0.01) + 0(0.99)\right]^2 \\
&= M^2 - \frac{M^2}{100} \\
&= \frac{99}{100}M^2
\end{aligned}$$

This is a HUGE variance, because $M = 10,000,000$. But for the most part, intuitvely, few people would say that this r.v. "varies" all that much. After all, it's almost always 0.

On the other hand, what if we compute the variance of Y? In this, we have $\mathbb{E}[Y] = -10\frac{1}{2} + 10\frac{1}{2} = 0$, so

$$\begin{aligned}
\text{Var}(Y) &= \mathbb{E}\left[Y^2\right] - 0 \\
&= (-10)^2\frac{1}{2} + (10)^2\frac{1}{2} \\
&= 100
\end{aligned}$$

In this case, the variance is pretty big—not huge, but pretty big. Yet the random variable Y doesn't really seem to "vary" all that much: after all, we know it is always either 10 more than the mean (which is zero), or 10 less than the mean.

End of Example 65. □

Hopefully this example doesn't give you the impression that variance is useless. It's actually a really important measure. The issue is just that random variables can "vary" in lots of different ways that can be hard to summarize as a single value.

Variance has several properties that are often useful, described in the next theorem.

Theorem 66 (Properties of Variance)**.** *Let X be a random variable. Then:*

- *For any scalars a, b,*
$$\mathrm{Var}(aX + b) = a^2 \mathrm{Var}(X). \tag{5}$$
- *If X and Y are independent random variables then*
$$\mathrm{Var}(aX + bY + c) = a^2 \mathrm{Var}(X) + b^2 \mathrm{Var}(Y). \tag{6}$$

Implications of Theorem 66:

- Taking $b = 0$ in Eq. (5) gives
$$\mathrm{Var}(aX) = a^2 \mathrm{Var}(X). \tag{7}$$

  Here is a direct proof of Eq. (7):

$$\begin{aligned}
\mathrm{Var}(aX) &= \mathbb{E}\left[(aX - \mathbb{E}[aX])^2\right] && \text{by Definition 62} \\
&= E\left[(aX - a\mathbb{E}[X])^2\right] && \text{by Proposition 59} \\
&= \mathbb{E}\left[a^2 (X - \mathbb{E}[X])\right] && \\
&= a^2 \mathbb{E}[X - \mathbb{E}[X]] && \text{by Proposition 59} \\
&= a^2 \mathrm{Var}(X) && \text{by Definition 62}
\end{aligned}$$

  □

- Taking $a = 1$ in Eq. (5) gives
$$\mathrm{Var}(X + b) = \mathrm{Var}(X)$$
  which says that adding a nonrandom quantity to a random variable doesn't change its variance.
- Taking $a = b = 1$ and $c = 0$ in Eq. (6) implies that
$$\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y).$$
  This says that for independent random variables, variance is additive.

  (Of course, we always have $\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$, even if X and Y are not independent. This is called ***linearity of expectation***. But for variance, we need X and Y to be independent to do something similar.)

15.2 Binomial random variable

Next we introduce an important type of random variable, called the "binomial" random variable. The motivation is the following idea: if you flip a coin n times, and let $\alpha \in (0, 1)$ be the probability of 'heads' on any given coin flip. We can define a random variable $X = (\#\text{ of heads in } n \text{ flips})$.

The probability that you get exactly k heads in n flips is given by the formula:
$$\binom{n}{k} \alpha^k (1 - \alpha)^{n-k}. \tag{8}$$

This is the probability that $X = k$. To see where Eq. (8) comes from, work out an example, say $n = 5$ and $k = 3$, and the general idea will become clear.

The precise definition of a binomial random variable is as follows:

Definition 67 (Binomial random variable)**.** A random variable is said to be a ***Binomial random variable*** with n trials and success probability α if

$$\mathbb{P}[X = k] = \binom{n}{k} \alpha^k (1-\alpha)^{n-k}$$

for every $k \in \{0, 1, \ldots, n\}$. In this case, we write $X \sim \text{Bin}(n, \alpha)$.

Remark 68 (Useful characterization of binomial random variables)**.** If X is a Binomial random variable with n trials and success probability α, then we can write

$$X = \xi_1 + \ldots + \xi_n$$

where

$$\xi_i = \begin{cases} 1 & : \text{ with probability } \alpha \\ 0 & : \text{ with probability } 1-\alpha \end{cases} .$$

In words ξ_i is either one or zero depending on whether the i^{th} coin flip was 'heads' or not. Hence, the sum $\xi_1 + \ldots + \xi_n$ counts the number of heads in n coin flips.

Example 69. Suppose we wish to compute $\mathbb{E}[X]$, when X is binomial with n trials and success parameter α.

To do this computation, we'll use the idea of Remark 68:

$$\begin{aligned} \mathbb{E}[X] &= \mathbb{E}[\xi_1 + \ldots + \xi_n] \\ &= \mathbb{E}[\xi_1] + \ldots + \mathbb{E}[\xi_n] && \text{by Proposition 59.} \end{aligned}$$

Observe that $\mathbb{E}[\xi_i] = 1 \cdot \alpha + 0 \cdot (1-\alpha) = \alpha$ for all $i \in \{1, \ldots, n\}$, and hence

$$\mathbb{E}[X] = n\alpha.$$

End of Example 69. □